Question # 7 of 10 (Start time: 09:34:42 PM, 30 August 2020) Total Marks: 1

The vector sum of $x = (x_1, x_2, x_3, x_4, x_5, x_6)$ and $y = (y_1, y_2, y_3, y_4, y_5, y_6)$ is $x + y =$ _____.

Select the correct option

Reload Math Equations

- $(x_1 + y_1 + x_2 + y_2 + \cdots + x_6 + y_6)$
- $(x_1 + y_1, x_2 + y_2, \cdots, x_6 + y_6)$
- $(x_1 + y_1, x_2 + y_2, \cdots, x_6 + y_6)$
- $(x_6 + y_6, x_5 + y_5, \cdots, x_1 + y_1)$

b

Time Left sec(s)

MTH631:quiz-3

Quiz Start Time: 09:26 PM, 30 August 2020

Question # 4 of 10 (Start time: 09:30:48 PM, 30 August 2020)

Total Marks: 1

In R^4 let $x = (1, -3, 5, 6)$ and $y = \left(2, -4, 3, \frac{1}{2}\right)$, $x + y =$ _____.

Select the correct option

Reload Math Equations

- $(3, -7, 8, 6.5)$
- $(2, -7, 8, 6)$
- $(3, 7, 8, 6.5)$
- $\left(3, -7, 8, \frac{7}{2}\right)$

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 09:32:02 PM, 30 August 2020)

Total Marks: 1

In the Schwarz's Inequality; $|X.Y| \leqslant |X|\,|Y|, \forall X, Y$ in R^n, the equlaity holds for - - - - - - - .

Select the correct option

Reload Math Equations

- $X = Y$
- $X = tY,\ t \in \mathrm{R}$
- $Y = tX,\ t \in \mathrm{R}$
- All above are valid

Question # 2 of 10 (Start time: 09:28:23 PM, 30 August 2020)

Total Marks: 1

If X and Y are in $\mathbb{R}^n$, then $|Y - X| - - - - -$.

Select the correct option

Reload Math Equations

- $\geqslant |Y| - |X|$
- $\leqslant |Y| - |X|$
- $\geqslant |X| - |Y|$
- $= |X| - |Y|$

Exterior of the set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is $----$.

Select the correct option

Reload Math Equations

- $\{(x,y) \in \mathbb{R}^2 \,|\, |x| < n, |y| > n, n \in \mathbb{N}\}$
- $\{(x,y) \in \mathbb{R}^2 \,|\, |x| > n, |y| < n, n \in \mathbb{N}\}$
- $\{(x,y) \in \mathbb{R}^2 \,|\, |x| > n, |y| > n, n \in \mathbb{N}\}$
- $\{(x,y) \in \mathbb{R}^2 \,|\, |x| < n, |y| < n, n \in \mathbb{N}\}$

Click to Save Answer & Move to Next Question

MTH631:quiz-3

Quiz Start Time: 09:14 PM, 30 August 2020

Question # 10 of 10 (Start time: 09:24:08 PM, 30 August 2020)

Total Marks: 1

Closure of the set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is $- - - -$.

Select the correct option

Reload Math Equations

- $\{(x,y) : -n < x, y < n, n \in \mathbb{N}\}$
- $\{(x,y) : -n \leq x, y \leq n, n \in \mathbb{N}\}$
- $\{(x,y) : (x,y) \neq (0,0)\}$
- $\{(x,y) : x, y \leq n, (x,y) \neq (0,0), n \in \mathbb{N}\}$

Question # 4 of 10 (Start time: 08:51:59 PM, 30 August 2020) Total Marks: 1

Both $X = X_0 + sV$ and $X = X_0 + tU$, $-\infty < s, t < \infty$ represent the same line through X_0 in $\mathbb{R}^n$, if - - - - .

Select the correct option

Reload Math Equations

○	$sV = tU$, $s \neq 0$ or $t \neq 0$
○	$V = \alpha U, \alpha \neq 0$
○	$U = \beta V$, $\beta \neq 0$
○	All above are equivalent

d

Question # 5 of 10 (Start time: 08:52:32 PM, 30 August 2020) Total Marks: 1

If X and Y are in $\mathbb{R}^n$, then $|X+Y|^2 \leq$ - - - - - -.

Select the correct option

Reload Math Equations

- $|X|^2 + 2(X.Y) + |Y|^2$
- $\left||X|^2 - |Y|^2\right|$
- $||X| - |Y||^2$
- $|X|^2 + 2|X||Y| + |Y|^2$

d

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 09:00:22 PM, 30 August 2020) **Total Marks:** 1

If $\bar{x}$ and $\bar{y}$ are in R^n and a is a scalar, then $(a\bar{x}) \cdot \bar{y} =$ _____.

Select the correct option

Reload Math Equations

- $\bar{x} \cdot (a\bar{y})$
- $a\,(\bar{x} \cdot \bar{y})$
- both a and b
- none of these

C

Question # 1 of 10 (Start time: 08:59:28 PM, 30 August 2020) Total Marks: 1

The length of the vector $\bar{x} = (x_1, x_2, 3, x_4, 5)$ is $|\bar{x}| =$ _____

Select the correct option

Reload Math Equations

- $(x_1^2 + x_2^2 + 9 + x_4^2 + 25)$
- $(x_1^2, x_2^2, 9, x_4^2, 25)^{\frac{1}{2}}$
- $(x_1^2 + x_2^2 + x_4^2 + 34)^{\frac{1}{2}}$
- $(x_1^2 + x_2^3 + 3^4 + x_4^5 + 5^6)^{\frac{1}{2}}$

Question # 6 of 10 (Start time: 08:53:30 PM, 30 August 2020) **Total Marks:** 1

$|\bar{x}|$ is the distance between x and _____.

Select the correct option

Reload Math Equations

- the origin
- $\bar{y}$
- x
- identity

MTH631:quiz-3

Quiz Start Time: 09:14 PM, 30 August 2020

Question # 8 of 10 (Start time: 09:21:31 PM, 30 August 2020)

Total Marks: 1

If the points 0, X, Y and $X+Y$ are joined to form a parallelogram in $\mathbb{R}^n$ of sides $|X|$ and $|Y|$, then its diagonal is represented by - - - - - - - .

Select the correct option

Reload Math Equations

- $|X|+|Y|$
- $|X|-|Y|$
- $|X+Y|$
- $||X|-|Y||$

Question # 9 of 10 (Start time: 09:10:53 PM, 30 August 2020)

Total Marks: 1

If $L: X = (-1, 3, -1) + t(2, -4, 0), -\infty < t < \infty$ is the equation of line through $(-1, 3, -1)$ in the direction of vector $(2, -4, 0)$ in R^3, then the line segment on L, when $t = 1$ is ------.

Select the correct option

Reload Math Equations

- $X = (1 - t)(-1, 3, -1) + t(2, -4, 0), 0 \leq t \leq 1$
- $X = t(-1, 3, -1) + (1 - t)(2, -4, 0), 0 \leq t \leq 1$
- $X = (1 - t)(-1, 3, -1) + t(1, -1, -1), 0 \leq t \leq 1$
- $X = t(-1, 3, -1) + (1 - t)(1, -1, -1), 0 \leq t \leq 1$

Question # 9 of 10 (Start time: 09:22:57 PM, 30 August 2020)

Total Marks: 1

Boundary of the set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is $- - - -$.

Select the correct option

Reload Math Equations

a

- $\{(x,y) \in \mathbb{R}^2 | x = \pm n, y = \pm n, n \in \mathbb{N}\} \cup \{(0,0)\}$
- $\{(x,y) \in \mathbb{R}^2 | x = \pm n, y = \pm n, n \in \mathbb{N}\}$
- $\{(0,0)\}$
- $\{(x,y) \in \mathbb{R}^2 | x = \pm n, y = \pm n, n \in \mathbb{N}\} \cap \{(0,0)\}$

Question # 6 of 10 (Start time: 09:07:31 PM, 30 August 2020)

Total Marks: 1

Both $X = X_0 + sV$ and $X = X_0 + tU$, $-\infty < s, t < \infty$ represent the same line through X_0 in $\mathbb{R}^n$, if - - - - .

Select the correct option

Reload Math Equations

- $sV = tU$, $s \neq 0$ or $t \neq 0$
- $V = \alpha U, \alpha \neq 0$
- $U = \beta V$, $\beta \neq 0$
- All above are equivalent

d

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 09:38:35 PM, 30 August 2020) Total Marks: 1

A sequence of points $\{X_r\}$ in R^n converges if and only if for each _____ there is an integer K such that _____ if $r, s \geq$ [redacted].

Select the correct option

Reload Math Equations

- $\varepsilon < 0, \quad |X_r - X_s| < \varepsilon$
- $\varepsilon > 0, \quad |X_r - X_s| > \varepsilon$
- $\varepsilon > 0, \quad |X_r - X_s| < \varepsilon$
- $\varepsilon \neq 0, \quad |X_r - X_s| > \varepsilon$

Question # 7 of 10 (Start time: 09:08:15 PM, 30 August 2020) Total Marks: 1

A sequence of points $\{X_r\}$ in R^n converges to the limit $\bar{X}$ if _____.

Select the correct option

Reload Math Equations

- lim
- \mathop {\lim }\limits_{r \to \infty } \left| {{X_r} - \bar X} \right| = 1
- \mathop {\lim }\limits_{r \to \infty } \left| {{X_r} - \bar X} \right| = 0
- \mathop {\lim }\limits_{r \to 0} \left| {{X_r} - \bar X} \right| = \bar X

Click to Save Answer & Move to Next Question

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MC190200352: IMRAN SHAHZAD

Time Left 81 sec(s)

MTH631:quiz-3

Quiz Start Time: 08:59 PM, 30 August 2020

Question # 4 of 10 (Start time: 09:02:08 PM, 30 August 2020) Total Marks: 1

The equation of a line through a point $X_0 = (x_0;\ y_0;\ z_0)$ in R^3 can be written parametrically as _____, $-\infty < t < \infty$ where $u_1, u_2, u_3 \neq 0$.

Select the correct option

Reload Math Equations

- $x = x_0 + u_1 t$
- $y = y_0 + u_2 t$
- $z = z_0 + u_3 t$
- all of above

d

Click to Save Answer & Move to Next Question

MTH631:quiz-3

Quiz Start Time: 09:14 PM, 30 August 2020

Question # 6 of 10 (Start time: 09:19:09 PM, 30 August 2020)

Total Marks: 1

The length of the vector $x = (x_1, x_2, 3, x_4, 5)$ is $|x| =$ _____

Select the correct option

Reload Math Equations

- $(x_1^2 + x_2^2 + 9 + x_4^2 + 25)$
- $(x_1^2, x_2^2, 9, x_4^2, 25)^{\frac{1}{7}}$
- $(x_1^2 + x_2^2 + x_4^2 + 34)^{\frac{1}{7}}$
- $(x_1^2 + x_2^2 + 3^4 + x_4^5 + 5^6)^{\frac{1}{7}}$

Question # 9 of 10 (Start time: 09:37:02 PM, 30 August 2020)

Total Marks: 1

If X and Y are in $\mathbb{R}^n$, then $|X+Y|^2 = - - - - - -$.

Select the correct option

Reload Math Equations

- $|X|^2 + 2|X||Y| + |Y|^2$
- $(|X| + |Y|)^2$
- $|X|^2 + 2(X.Y) + |Y|^2$
- $X^2 + 2|X||Y| + Y^2$

Click to Save Answer & Move to Next Question

MTH631:quiz-3

Quiz Start Time: 09:14 PM, 30 August 2020

Question # 5 of 10 (Start time: 09:17:35 PM, 30 August 2020)

Total Marks: 1

If X and Y are in $\mathbb{R}^3$, *then* $|X-Y|$ ----.

Select the correct option

Reload Math Equations

- $\leqslant ||X|-|Y||$
- $\leqslant |X|-|Y|$
- $\geqslant ||X|-|Y||$
- $\leqslant |Y|-|X|$

Question # 3 of 10 (Start time: 09:29:41 PM, 30 August 2020)

Total Marks: 1

If X and Y are in $\mathbb{R}^n$, then $|X+Y|^2 \leqslant -----$.

Select the correct option

Reload Math Equations

- $|X|^2 + 2(X.Y) + |Y|^2$
- $\left||X|^2 - |Y|^2\right|$
- $||X| - |Y||^2$
- $|X|^2 + 2|X||Y| + |Y|^2$

d

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 09:17:04 PM, 30 August 2020) Total Marks: 1

The set $\{(x, y) : -n < x, y < n, (x, y) \neq (0, 0), n \in \mathbb{N}\}$ is - - - - - - in $\mathbb{R}^2$.

Select the correct option

Reload Math Equations

- open
- closed
- neither open nor closed
- both open or closed

Question # 9 of 10 (Start time: 08:55:42 PM, 30 August 2020)

Total Marks: 1

If X_0 is a point in R^n and r is a positive number, the open n-ball of radius r about X_0 is the set $B_r(X_0) = \{X| \ |X - X_0| _____ \}$

Select the correct option

Reload Math Equations

- $\leq r$
- $< r$
- $\geq r$
- $> r$

Question # 7 of 10 (Start time: 09:08:15 PM, 30 August 2020) Total Marks: 1

A sequence of points $\{X_r\}$ in R^n converges to the limit $\bar{X}$ if _____

Select the correct option

Reload Math Equations

- $\lim_{r \to 0} |X_r - \bar{X}| = 0$
- $\lim_{r \to \infty} |X_r - \bar{X}| = 1$
- $\lim_{r \to \infty} |X_r - \bar{X}| = 0$
- $\lim_{r \to 0} |X_r - \bar{X}| = \bar{X}$